

Exercises V : Topology III

October 30, 2025

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1. **a)** Compute the homology groups of $Gr_2(\mathbb{C}^4)$.
b) Prove that the map $Gr_n(\mathbb{C}^m) \rightarrow Gr_n(\mathbb{C}^\infty)$ is a $2(m - n) - 1$ -equivalence.
 2. Let $W_k(\mathbb{C}^n)$ denote the (complex) Stiefel manifold of k -orthonormal vectors in $\mathbb{C}^n$. Prove that $H^*(W_k(\mathbb{C}^n)) \cong \Lambda(y_{2n-2k+1}, y_{2n-2k+3}, \dots, y_{2n-1})$, an exterior algebra with odd degree generators y_{2j+1} in degree $2j + 1$. In particular, $H^*(U(n)) \cong \Lambda(y_1, y_3, \dots, y_{2n-1})$.
 3. Compute all the Steenrod operations on $H^*(W_k(\mathbb{C}^n); \mathbb{Z}/2)$ and $H^*(U(n); \mathbb{Z}/2)$.
 4. Suppose that $F \rightarrow E \rightarrow X$ is a fibration with X simply connected and both H^*F (assume also that H^*F has finite rank), H^*X torsion free and concentrated in even degrees (that is, the cohomology groups in odd degree are 0). Prove that the fibration satisfies the hypothesis of Leray-Hirsch theorem.
 5. Hatcher Sec 4.D 1-4.
 6. a) Let ξ be a complex vector bundle over a space X . Prove that $P(\xi)$, the projectivisation of ξ , satisfies the hypothesis of the Leray-Hirsch theorem.
b) For a real vector bundle, prove that the same consequence holds with $\mathbb{Z}/2$ -coefficients.
 7. Consider $U(n-1) \subset U(n)$. Prove that the canonical n -plane bundle γ_n pulls back to $\gamma_{n-1} \oplus \epsilon_{\mathbb{C}}$ over $BU(n-1)$. Also prove that $BU(n-1) \simeq S(\gamma_n)$, the sphere bundle of γ_n .
 8. Let X be a CW complex of dimension $\leq 2n - 1$, and let ξ be a complex vector bundle of dimension n over X . Prove that there is a complex vector bundle ξ' such that $\xi \cong \xi' \oplus \epsilon_{\mathbb{C}}$.
 9. Let X be a CW complex of dimension $\leq 2d - 1$, and let ξ be a complex vector bundle of dimension n over X . Prove that there is a complex vector bundle ξ' dimension d such that $\xi \oplus \xi'$ is a trivial bundle.